

Test-case results across three Isaac Sim roads.

Method	Tested map not in D_{cal} ?	icySide ID	icySide OOD	icyMain ID	icyMain OOD	icyMiddle ID	icyMiddle OOD
Marginal coverage (%) ↑							
NoCP	N/A	90.0	56.7	90.0	56.7	90.0	56.7
SplitCP	×	99.5	89.6	100.0	98.4	99.1	85.9
LUCCa	×	91.1	91.5	90.1	91.4	90.1	90.9
OCULAR (ours)	✓	91.5	90.1	90.4	90.1	91.1	90.6
Median volume (relative to oracle) ↓							
NoCP	N/A	1.00	0.28	1.00	0.28	1.00	0.28
SplitCP	×	3.73	1.03	8.68	2.40	3.07	0.85
LUCCa	×	1.08	1.13	1.02	1.10	1.02	1.13
OCULAR (ours)	✓	1.03	1.02	1.02	1.02	1.06	1.06

red means coverage below 90%. Prediction region volume ratio is reported relative to an oracle using the minimum scaling factor needed to achieve 90% coverage. Each map has 4464 test transitions.